

SOROCHINSK, Russian Federation

WMO: 350110

Lat: 52.433N

Lon: 53.133E

Elev: 123

StdP: 99.86

Time Zone: 5.00 (E05)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.9	-24.8	-31.1	0.2	-27.8	-27.8	0.3	-24.3	8.5	-5.2	8.0	-5.6	0.9	120	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	34.7	19.0	32.7	18.4	31.0	18.0	20.4	29.8	19.7	29.0	19.0	28.5	3.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	12.9	22.7	16.7	12.1	22.5	15.7	11.3	22.1	59.1	29.8	56.5	28.7	54.3	28.6	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.1	7.0	6.2	-31.3	36.8	3.6	1.2	-33.9	37.7	-35.9	38.3	-37.9	39.0	-40.5	39.8
			-31.5	21.5	3.5	0.7	-34.0	22.0	-36.0	22.4	-38.0	22.7	-40.5	23.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-12.6	-11.4	-4.5	7.5	16.0	20.4	22.4	21.0	14.3	5.9	-1.3	-8.2	(6)
(7)		DBStd	13.54	6.51	6.39	5.23	5.57	4.30	4.55	3.35	4.76	4.32	4.74	5.16	6.32	(7)
(8)		HDD10.0	2926	701	600	448	114	4	1	0	1	12	144	339	563	(8)
(9)		HDD18.3	4918	959	833	707	326	98	30	6	28	133	386	589	822	(9)
(10)		CDD10.0	1422	0	0	0	39	191	312	384	341	140	16	0	0	(10)
(11)		CDD18.3	372	0	0	0	2	27	91	132	110	11	0	0	0	(11)
(12)		CDH23.3	4423	0	0	0	25	356	1118	1531	1261	132	0	0	0	(12)
(13)		CDH26.7	1836	0	0	0	2	91	492	644	579	28	0	0	0	(13)
(14)	Wind	WSAvg	2.8	2.8	2.6	3.1	3.1	2.8	2.7	2.4	2.4	2.5	2.9	3.0	2.9	(14)
(15)	Precipitation	PrecAvg	410	30	23	29	28	35	51	39	39	32	36	36	32	(15)
(16)		PrecMax	589	56	48	70	88	95	141	92	96	92	79	92	76	(16)
(17)		PrecMin	294	6	6	2	4	2	1	1	4	2	3	4	6	(17)
(18)		PrecStd	76	12	10	17	20	24	34	28	24	22	19	25	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.6	1.4	10.6	26.5	31.0	35.5	36.5	36.8	29.9	20.6	10.5	3.2	(19)
(20)			MCWB	-0.1	0.6	6.5	14.3	16.3	18.8	19.9	19.1	15.8	11.3	8.5	2.6	(20)
(21)		2%	DB	-0.4	0.4	4.5	23.2	29.2	33.8	34.4	34.5	27.0	17.5	8.4	1.8	(21)
(22)			MCWB	-0.9	-0.2	2.2	13.0	15.9	18.4	19.7	18.7	15.2	10.3	6.6	1.1	(22)
(23)		5%	DB	-1.7	-0.7	3.0	19.8	27.3	32.0	32.2	32.3	24.5	15.1	6.7	0.9	(23)
(24)			MCWB	-2.2	-1.4	1.3	11.4	15.4	17.8	18.9	18.3	14.2	9.2	5.2	0.3	(24)
(25)		10%	DB	-3.5	-2.4	1.8	16.8	25.2	29.7	30.3	30.2	22.2	12.8	5.1	0.1	(25)
(26)			MCWB	-4.1	-3.3	0.5	9.7	14.6	17.6	18.3	17.9	13.6	8.4	4.1	-0.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	0.9	6.6	15.2	17.6	20.6	21.7	20.7	17.7	13.2	9.1	2.7	(27)
(28)			MCDWB	0.6	1.2	10.3	24.2	27.0	30.9	31.6	29.5	26.4	17.2	10.1	3.0	(28)
(29)		2%	WB	-0.8	-0.1	2.6	13.7	16.8	19.7	20.9	19.8	16.3	11.2	6.8	1.2	(29)
(30)			MCDWB	-0.4	0.3	4.0	21.5	26.2	29.6	30.5	29.4	23.2	15.3	8.0	1.5	(30)
(31)		5%	WB	-2.2	-1.3	1.5	12.1	16.1	19.0	20.0	19.2	15.5	10.0	5.4	0.5	(31)
(32)			MCDWB	-1.8	-0.7	2.6	18.3	25.7	28.5	29.0	29.3	21.8	13.3	6.4	0.9	(32)
(33)		10%	WB	-4.0	-3.3	0.7	10.5	15.1	18.4	19.2	18.6	14.4	9.0	4.2	-0.6	(33)
(34)			MCDWB	-3.5	-2.5	1.7	16.4	23.6	27.2	28.0	28.3	20.5	12.2	5.1	0.0	(34)
(35)	Mean Daily Temperature Range	MDBR		6.5	7.5	7.4	10.0	12.0	12.4	12.6	12.1	10.4	7.8	4.7	5.5	(35)
(36)		5% DB	MCDBR	5.7	5.3	6.8	14.5	15.2	15.3	15.1	14.6	14.7	11.9	5.3	4.0	(36)
(37)			MCWBR	5.8	4.9	5.2	6.7	5.5	5.0	4.9	4.4	5.9	6.3	4.2	4.1	(37)
(38)			MCDBR	5.8	4.6	6.3	12.7	13.9	12.7	12.8	13.2	11.8	9.5	4.8	4.0	(38)
(39)		5% WB	MCWBR	5.8	4.5	5.1	6.4	5.7	4.8	4.7	4.5	5.4	5.7	4.4	4.1	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.302	0.327	0.377	0.433	0.409	0.402	0.409	0.428	0.382	0.351	0.330	0.302	(40)
(41)		taud		2.057	2.020	1.963	2.063	2.250	2.326	2.306	2.234	2.324	2.319	2.223	2.077	(41)
(42)		Ebn at Noon		646	750	786	795	842	851	837	795	791	735	624	570	(42)
(43)		Edh at Noon		78	109	143	147	128	120	121	123	100	82	67	64	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.08	2.21	3.62	4.53	5.73	6.25	6.10	5.09	3.45	1.78	0.79	0.66	(44)
(45)		RadStd		0.15	0.16	0.41	0.49	0.45	0.64	0.52	0.42	0.46	0.30	0.14	0.16	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	+1.00	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page